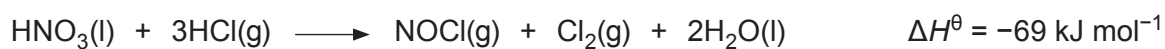


10. Aqua regia is an acidic liquid that can dissolve unreactive metals such as gold. It can be formed by bubbling hydrogen chloride gas through concentrated nitric acid.



Substance	Standard enthalpy change of formation, $\Delta_f H^\theta / \text{kJ mol}^{-1}$	Standard entropy, $S^\theta / \text{J K}^{-1} \text{mol}^{-1}$
$\text{HNO}_3(\text{l})$	-173	156
$\text{HCl}(\text{g})$		187
$\text{NOCl}(\text{g})$	53	264
$\text{Cl}_2(\text{g})$	0	223
$\text{H}_2\text{O}(\text{l})$	-286	70

- (a) Show that this is a redox reaction. [1]

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- (b) State why the enthalpy change of formation of $\text{Cl}_2(\text{g})$ is zero. [1]

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.....

- (c) Calculate the standard enthalpy change of formation of $\text{HCl}(\text{g})$. [2]

$$\Delta_f H^\theta = \dots\dots\dots \text{kJ mol}^{-1}$$



- (d) (i) Calculate the free energy change, $\Delta G^\ominus$, for this reaction at 25 °C.

[3]

$$\Delta G^\ominus = \dots\dots\dots \text{kJ mol}^{-1}$$

- (ii) An alternative method of preparing aqua regia is to use concentrated hydrochloric acid with concentrated nitric acid. State, giving a reason, the effect of this on the value of $\Delta G^\ominus$ at 25 °C.

[1]

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- (e) (i) The pH of a dilute solution of nitric acid is measured using a pH probe and has a value of 0.3. Calculate the concentration of this solution.

[1]

$$\text{Concentration} = \dots\dots\dots \text{mol dm}^{-3}$$

- (ii) An alternative way of measuring the concentration of the solution is by titration against a standard solution of sodium hydrogencarbonate of concentration 0.500 mol dm⁻³.

State and explain which of these two methods will give the more precise value for the concentration.

[2]

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10. (a) (i) Tin and carbon are both more stable in oxidation state +4 whilst lead is more stable as +2. Explain the difference in the stabilities of these oxidation states. [1]

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- (ii) Lead(IV) oxide can act as an oxidising agent when it reacts with concentrated hydrochloric acid. Write an equation for this reaction. [1]

.....

- (b) Tin(II) ions can be used to reduce iron(III) ions in **acidic** solution.



$[\text{Fe}^{3+}] / \text{mol dm}^{-3}$	$[\text{Sn}^{2+}] / \text{mol dm}^{-3}$	pH	Initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
0.200	0.100	0	1.4×10^{-2}
0.300	0.200	0	2.8×10^{-2}
0.400	0.100	1	1.4×10^{-3}
0.400	0.200	1	2.8×10^{-3}

- (i) Find the rate equation for this reaction. [3]

- (ii) Calculate the value of the rate constant, k , for this reaction. [2]

$k =$



- (iii) A student suggests that the rate determining step for this reaction is:



State, giving a reason, whether this is a possible rate determining step for this reaction. [2]

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- (c) The rate of this reaction can be followed using sampling and quenching.

- (i) Explain what is meant by *sampling and quenching*. [1]

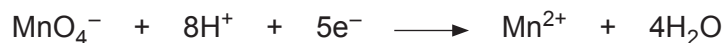
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- (ii) A 5.00 cm^3 sample of the solution was analysed by titration against acidified potassium manganate(VII). The sample required 27.20 cm^3 of manganate(VII) solution of concentration $0.00205\text{ mol dm}^{-3}$ for complete reaction.

Calculate the concentration of Fe^{2+} in the solution. [2]



Concentration of $\text{Fe}^{2+} = \dots\dots\dots \text{mol dm}^{-3}$



10. Iodomethane, CH_3I , reacts with a range of nucleophiles. The rate of its reaction with thiosulfate ions, $\text{S}_2\text{O}_3^{2-}$, has been studied at a range of temperatures.

(a) The reaction is found to be first order with respect to iodomethane and has a rate constant of $3.10 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ at a particular temperature.

(i) Write the rate equation for this reaction. [2]

.....

(ii) The activation energy for this reaction is 81.1 kJ mol^{-1} with a frequency factor, A , of $5.62 \times 10^{12} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$. Find the temperature used for this reaction. [3]

$T = \text{..... K}$



(b) One method of performing this experiment involves sampling and quenching.

(i) Explain what is meant by the term 'quenching' and why it needs to be carried out. [2]

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(ii) One method of measuring the concentration of thiosulfate ions in solution is by titration using a solution containing iodine. Suggest an indicator for this titration. [1]

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(iii) A student suggests using gravimetric analysis to measure the amount of iodide ions produced in the reaction by addition of $\text{Pb}^{2+}(\text{aq})$ ions to form a precipitate. Give the colour of this precipitate. [1]

.....

(c) Iodomethane is an intermediate in the conversion of methanol to ethanoic acid.

(i) One catalyst used for this reaction is the iridium-containing complex ion $[\text{Ir}(\text{CO})_2\text{I}_2]^-$. It is an example of a homogeneous catalyst.

I. State what is meant by a *homogeneous* catalyst. [1]

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II. Explain why transition metal complexes such as $[\text{Ir}(\text{CO})_2\text{I}_2]^-$ can act as homogeneous catalysts. [2]

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- (ii) Ethanoic acid is a weak acid with $K_a = 1.76 \times 10^{-5} \text{ mol dm}^{-3}$. Calculate the pH of an aqueous solution of ethanoic acid with a concentration of $0.220 \text{ mol dm}^{-3}$. [3]

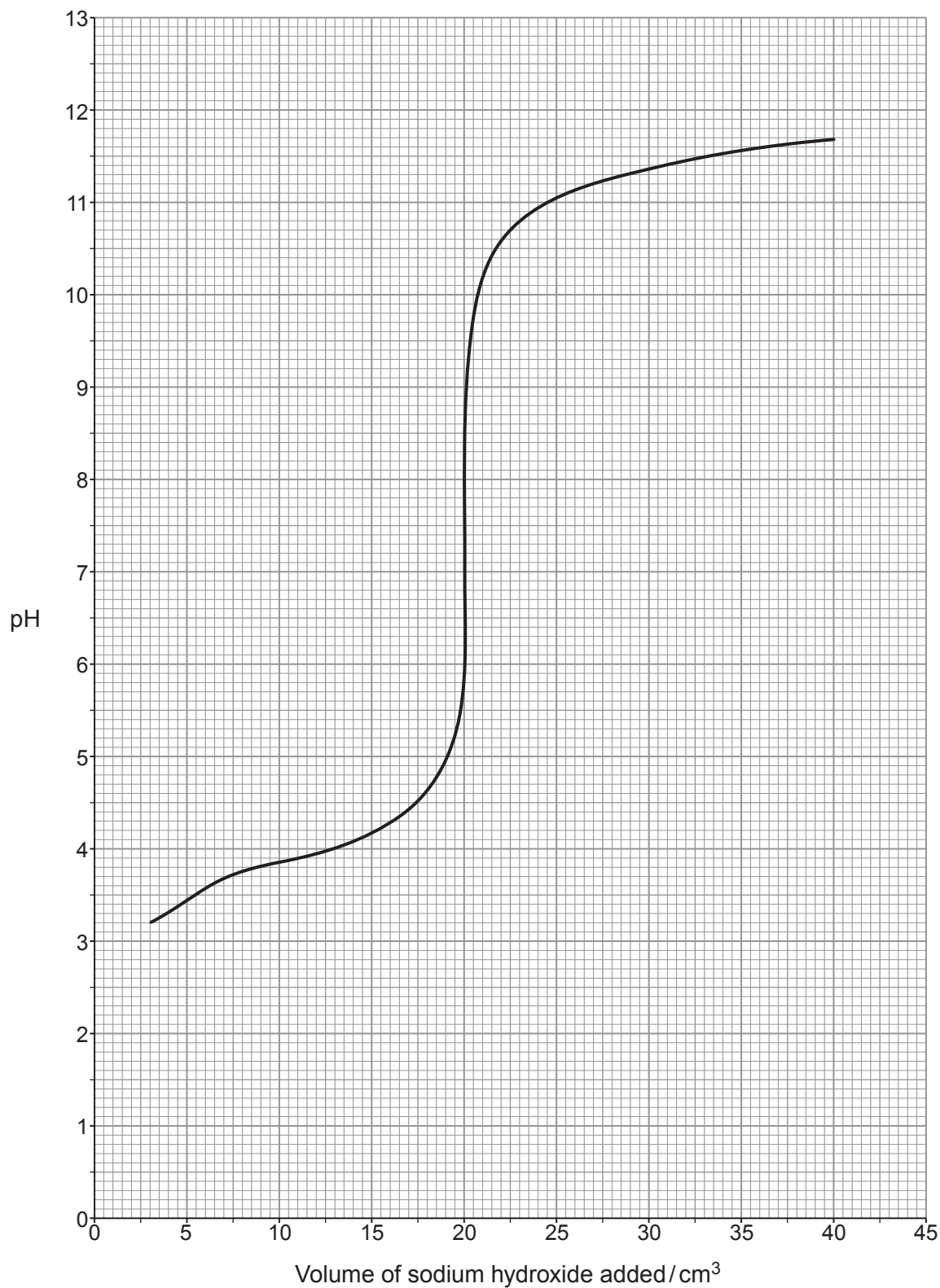
pH =

Examiner
only

15



10. A student used a solution of weak acid HA in a titration. A 25.0 cm^3 sample of the acid was titrated against a sodium hydroxide solution of concentration 0.250 mol dm^{-3} giving the pH titration curve shown. The initial pH value is missing from the graph.



- (a) Several indicators are listed in the table.

Identify **all** the indicator(s) which are appropriate for this titration. Give a reason for your choice(s).

[2]

Indicator	pH range
methyl orange	3.1 - 4.4
bromocresol purple	5.2 - 6.8
bromothymol blue	6.0 - 7.6
naphtholphthalein	7.3 - 8.7
cresolphthalein	8.2 - 10.1

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- (b) Calculate the concentration of the HA solution.

[2]

Concentration = mol dm^{-3}



- (c) Find the pH of the mixture after addition of 10.0 cm^3 of sodium hydroxide solution and hence calculate the initial pH of the HA solution. [4]

You **must** show your working.

pH after addition of 10.0 cm^3 =

Initial pH =

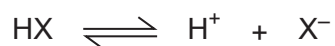
- (d) HA is a weak acid and so it can be used to make a buffer solution.

Suggest **one** important use for buffer solutions. [1]

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- (e) Some data regarding the dissociation of another weak acid, HX, at two different temperatures are given in the table.



	HX at 298 K	HX at temperature T
Percentage dissociated	3.9	3.4
ΔG for acid dissociation / kJ mol^{-1}	15.9	16.7
ΔH for acid dissociation / kJ mol^{-1}	-10.0	
ΔS for acid dissociation / $\text{J K}^{-1} \text{mol}^{-1}$	-87.0	

Find temperature T and hence explain in terms of equilibrium why the percentage dissociation is different at this temperature.

[3]

$T = \dots\dots\dots \text{K}$

.....



10. Oxygen forms several molecular compounds of formula X_2O . These include compounds with fluorine and chlorine.

(a) Some thermodynamic data for F_2O and Cl_2O are given below.

Substance	Standard enthalpy change of formation, $\Delta_f H^\theta / \text{kJ mol}^{-1}$	Standard molar entropy, $S^\theta / \text{JK}^{-1} \text{mol}^{-1}$
F_2O	24.5	247
Cl_2O	80.3	266
O_2		205
F_2		203
Cl_2	0	

A student examines the data and states that there is sufficient information to find the minimum temperature for decomposition of F_2O to its elements but not for Cl_2O .

Is the student correct? Justify your answer.

[3]

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- (b) F_2O is a strong oxidising agent and can oxidise Xe to XeF_4 whilst releasing oxygen gas.

(i) Write an equation for this process.

[1]

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(ii) Show that the xenon is oxidised in the process.

[1]

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- (c) F_2O reacts slowly with water to produce HF, a weak acid.



A small amount of F_2O is added to 400 cm^3 of water at 15°C and it reacts completely. The reaction produces 82 cm^3 of oxygen gas at this temperature and 1 atm pressure.

- (i) Calculate the concentration of the HF solution formed. [2]

concentration = mol dm^{-3}

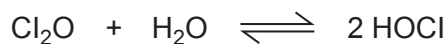
- (ii) Calculate the pH of the HF solution formed. [3]

$$K_a(\text{HF}) = 6.6 \times 10^{-4} \text{ mol dm}^{-3}$$

pH =



- (d) Cl_2O can react with water in a reversible reaction.



This reaction can occur in non-aqueous solvents and in a particular solvent the value of the equilibrium constant, K_c , at 273 K is 5.0 and it has no units.

- (i) Write an expression for the equilibrium constant K_c . [1]

- (ii) A student studies the equilibrium at a higher temperature. She adds samples of 0.40 mol of H_2O and 0.14 mol of Cl_2O to the non-aqueous solvent giving a total volume of 1000 cm^3 .

At equilibrium the concentration of HOCl is 0.22 mol dm^{-3} .

- I. Find the value of the equilibrium constant K_c at this temperature. [3]

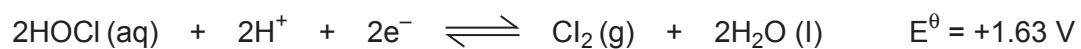
$K_c = \dots\dots\dots$

- II. State, giving a reason, whether this reaction is endothermic or exothermic. [2]

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III. HOCl can act as an oxidising agent.



Give the formulae of all the metal ions in the table that can be oxidised by HOCl (aq). Explain why you have chosen these ions. [2]

	Standard electrode potential, E^θ/V
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	+0.77
$\text{Ti}^{3+} + 2\text{e}^- \rightleftharpoons \text{Ti}^+$	+1.25
$\text{Pb}^{4+} + 2\text{e}^- \rightleftharpoons \text{Pb}^{2+}$	+1.69
$\text{Co}^{3+} + \text{e}^- \rightleftharpoons \text{Co}^{2+}$	+1.82

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